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"Introducing a *Platform* for Full Sky Kinetics and Imaging [FSKI]"

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ABSTRACT

The FSKI platform enhances the current imaging technology which is constrained by the Field of View (FoV) provided by the state of the art CCD designs and the associated astrometry, photometry and pipelines for delivering images. Even with the advanced field of views enabled by the Rubin Observatory (https://rubinobservatory.org/explore/how-rubin-works/lsst) and Nancy Grace Roman telescope (https://science.nasa.gov/mission/roman-space-telescope/) the point-and-image strategy remains the default technique of our time. Specify a given Right Ascension (RA) and Declination (Dec), then build a plan, schedule, initiate, slew, acquire image, readout, and downlink the data to Earth for more pipeline processing. This repeats for each RA and Dec target. The ability to further the strategy to leverage time domain astronomy employs a repetition of the process between time deltas in order to task the pipeline with characterizing the differences.

Keywords: Celestial sphere, Field of View (Fov), Charged Couple Device (CCD), CCD readout, slewing, downlink, pipeline, High-gain antenna (HGA), National Institude of Standards (NIST), Rubin Observatory (LSST), Nancy Grace Roman telescope, Right Ascension (RA), Declination (Dec), Astronomical Unit (AU), Lagrange points, Goldberg polyhedron, James Webb Space Telescope (JWST), payload fairing (PLF), celestial sphere, horseshoe orbits, co-orbital configuration, stereo orbits, superior conjunctive orbits

1. PROPOSAL

The science and technology of today suggests no options to radically expand the ability from the RA/Dec point-and-image strategy to larger FoV's. More aggregation of image collection combined with a larger FoV can lead to full sky coverage, if designed properly. This document composes a **platform design** that enables Full Sky Kinematics and Imaging (FSKI). The FSKI success depends on optimal deployment in outer space or terrestially. The correlative technology will incorporate imaging and state-of-the-art pipeline processing.

2. DO THIS FIRST

To fully appreciate this pdf document along with the links/references to the figure images and the movie .mp4 animations, clone the GitHub repository to a local workstation directory. For complete details on cloning the FSKIDesignAndSupplemental GitHub repository refer to THE FSKIDESIGNANDSUPPLEMENTAL GITHUB REPOSITORY in the appendix.

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3. CRITICAL ENABLING FACTORS

The implied goal throughout this document is full sky coverage.

The two prominent critical enabling factors for this goal of full sky coverage with FSKI:
- Ultimate configuration of the FSKI platform
- Deployment for regular operations of the FSKI platform

4. FSKI PROGRESSIVE SERIES OF FIGURES

Is there a strategy that would build upon current architectures with the goal of *full* sky coverage in tandem with a cadence that enables full sky time-domain astronomy?

The intent of this document is to suggest a means to accomplish full sky/sphere coverage with FSKI. Despite the lack of larger FoV's, this capability can still aspire to cover the full sky and enable time-domain capabilities.

In order to explain the platform design and deployment, this document will employ a progressive series of figures in the appendix to permit full appreciation of the end-state vision.

- Full Goldberg polyhedron{3,3}[1] with 12 pentagons and 260 hexagons in total
- Basic truncated polyhedron
- Tandem pair of truncated polyhedrons
- Vacant spaces within the truncated polyhedrons (for outer space deployment)
- Interior core to serve as equipment bay
- Full configurations for outer space operations with added support post, solar radiation shield and high gain antenna
- Abbreviated view of tandem pair for deployment in solar orbit
- One possible embodiment of example lens distribution on a hexagon face with n lenses per face
- Lens configurations on the face of a hexagon
- Terrestrial FSKI platform (with animation)
- L2, L3, L4 deployment (with animation)
- L4, L5 deployment (with animation)
- L4, L5 deployment (with animation) with tandem pair at L4 and L5 locations
- Example 1/2 sphere sky coverage at 2.50 degree target separation
- Example 1/2 sphere sky coverage at 1.25 degree target separation

Please refer to the Appendix Figures B1 - B17.

5. GITHUB AND PYTHON JUPYTER LAB PROJECT REPOSITORY

The GitHub repository is at: (https://github.com/fklich/FSKIDesignAndSupplemental)

The Jupyter Notebook python cells generate animation files (.mp4) to illustrate platform orbital deployment. Also, two 1/2 sphere sky coverage ICRS maps are plotted for two target *degrees of separation*. This document will cite specific animation .mp4 files located in the repository as necessary. Readers can download and play the files, complete with sound effects. For complete details on cloning the *FSKIDesignAndSupplemental* GitHub repository refer to *THE FSKIDESIGNANDSUPPLEMENTAL GITHUB REPOSITORY* in the appendix.

6. REFERENCES

Oripov, B. G. and Rampini, D. S. and Allmaras, J. and Shaw, M. D. and Nam, S. W. and Korzh, B. and McCaughan, A. N. (2023). A superconducting nanowire single-photon camera with 400,000 pixels. arXiv:2306.09473. , (https://arxiv.org/abs/2306.09473)

Klich, F. S., (2026). Full Sky Kinematics and Imaging Platform. United States Patent and Trademark Office, Application Number **64/072, 112**, Filed on 05/22/2026@11:43:20 AM Z ET, Provisional Patent Application

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Klich, F. S., (2026). Current resume in both .docx format and .pdf format available as download from GitHub, (https://github.com/fklich/FredKlichResume)

7. POLYHEDRON DESIGN AND FREQUENCY

The frequency represents the number of tapered chambers, the locations of the imaging technology on board the FSKI platform. Each chamber can/would accomodate 1-n lenses and telescope apparati. For the sake of an illustration of the platform design a Goldberg polyhedron{3,3} is depicted (https://en.wikipedia.org/wiki/Goldberg_polyhedron). This construct will yield 12 pentagons and 260 hexagons for the fully populated polyhedron. This polyhedron is further truncated to 182 faces as shown in the FSKI progressive series of figures section.

8. PHYSICAL SIZE OF THE POLYHEDRON

A rough estimate of the polyhedron diameter can be suggested by the dimensions of the payload fairing (PLF) used for the James Webb Space Telescope launch atop the Ariane 5 ECA+ rocket (https://en.wikipedia.org/wiki/Ariane_flight_VA256) . This is an arbitrary reference for the sake of common sense sizing. If the assumption is that the PLF will use a long payload fairing, then this compartment should accomodate two spheres that are 15 feet (4.6m) in diameter, placed atop each other. Likely, this payload configuration would be judged by the orbital deployment (i.e. L2, L3 or L4, L5). This broad sketch does suggest that the FSKI would **not** necessitate a gigantic assembly *and* might conveniently fit into a proven payload compartment. Also, the image acquisition design as facilitated by the lens size and layout strategy atop each tapered chamber could easily mitigate the required hexagonal facial area. A smaller facial area suggests smaller overall polyhedron diameter.

9. SKY COVERAGE

This will be dictated by the 1-n lens layouts atop each tapered chamber. The FSKI design is targeted as a

[1] A Goldberg polyhedron{3,3} denoting a regular polyhedron consisting of 3-sided polygon faces, with 3 faces joining at each vertex

platform for a *survey-class* instrument. Each individual telescope, lens assembly and its associated imaging equipment for this platform is bracketed between 2-4 square degrees and possibly narrower. Again, the factors governing the sky coverage is N lenses (total lenses) incorporated throughout all chambers, the FoV against 1/2 of the celestial sphere, 1/2 of 41,253 square degrees. The other prominent driver in sky coverage estimation is whether more than one tandem pair of FSKI's are deployed at the Lagrange point.

10. MORE THAN ONE POLYHEDRON PAIR

This original design constrained the deployment to one pair of polyhedrons, deployed at L2 and L3, using an intermediate relay satellite. An additional pair of polynedrons can be situated at L4 and L5 locations as suggested in the FSKI progressive series of figures section.

11. LAGRANGE LOCATIONS

The Lagrange points are suggested for FSKI deployment within the solar system. (https://en.wikipedia.org/wiki/Lagrange_point) , abbreviated as L2 and L3. L2 and L3 are opposite the sun. L4 and L5 are approximately 60 degrees ahead and behind the Earth orbital track.

12. L2 AND L3 DEPLOYMENT

Refer to Figure B11. This shows the outward facing FSKI platform covering the right side celestial sphere and left side celestial sphere (in general terms). Spherical overlap is intentional in the platform design. Since each FSKI platform exists as sun-opposites (conjunctive orbits), an L4 relay satellite is depicted. The overlapping band between the two platforms is a circular shape roughly centered on the sun. The GitHub resources include an .mp4 file that simulates the orbital dynamics of this option. This animation is embedded in Figure B11 and can be downloaded from (https://github.com/fklich/FSKIDesignAndSupplemental/blob/main/FSKIOrbitAnimationAtL2L3L4.mp4) including an audio soundtrack. Click download on right side.

13. L4 AND L5 DEPLOYMENT

Refer to Figure B12. This shows the deployment at the stable L4 and L5 points. While these two locations will NOT be direct solar opposites, still, the full celestial sphere indeed is covered. The overlaping band between the two platforms is an elliptical shape. The GitHub resources include an .mp4 file that simulates the orbital dynamics of this option. This animation is embedded in Figure B12 and can be downloaded from (https://github.com/fklich/FSKIDesignAndSupplemental/blob/main/FSKIOrbitAnimationAtL4L5.mp4) including an audio soundtrack. Click download on right side.

14. TERRESTRIAL DEPLOYMENT, POLES, HEMISPHERES, ETC.

Using the FSKI platform for terrestrial observation is feasible. The sphere would face the nighttime celestial dome likely elevated on a shaft to leverage horizon-to-horizon access. For terrestrial use the polyhedron will have *no* vacant chambers, facilitating horizon-to-horizon coverage as the FSKI tracks sky movement east to west. The sphere would be integrated with a mount to follow the sky motion, like a conventional equatorial telescope.

Consideration can be lent to latitude/longitude locations. Perhaps terrestrial polar locations would have some advantages with regard to winter skies and minimal Solar radiation interference.

The readout and downlink functions might function the same as an outer space deployment, even using radio transmission to deliver the raw image data for pipeline processing.

Another perspective on Terrestrial deployment is that it enables two tiers of sky exploration; (1) the Earth sky dome (i.e. day or night) and (2) the full celestial space dome. Surveys of Earth daytime might not necessitate any measure of *tracking*, just static images.

This animation is embedded in Figure B10 and can be downloaded from (https://github.com/fklich/FSKIDesignAndSupplemental/blob/main/FSKITerrestrialEmbodiment.mp4) including an audio soundtrack. Click download on right side.

15. ON BOARD ARCHITECTURES FOR FSKI

The FSKI will incorporate proven technologies for operation on Earth and in space.

- Power bus
- Attitude Control Subsystem (ACS)
- Fuel for station keeping
- Ion thrusters
- Solid State Recorder for onboard data storage
- Radios
- Ka-Band High Gain antenna to transmit science images
- S-Band Medium Gain Antenna for equipment commands and software maintenance
- Cryo supply for cryogenic cooling
- Cameras, image export equipment and software
- Solar shield

- Solar arrays for onboard power supply

16. TILTING STRATEGY AMONGST POLYHEDRONS

FoV coverage can be a strategy predicated on the number of lenses n and the number of FSKI's deployed. Each **tandem pair (TP)** can be tilted in contrast to another matched-partner **tandem pair (TP)** to aim the 1-n lenses at a unique **target set** in the sky.

For example if the lens set in hexagon cell #45 points directly at coordinate (RA, Dec) on TPa, then cell #45 on the associated tandem twin could be aimed at TPa' (RA+1 degree, Dec+1 degree), tilted by 1 degree between the twin FSKI platforms. This tilting strategy could expand to even more matched tandem pairs for a more expansive blanket/tapestry FoV's. The FSKI's can be separated by 200 kilometers and not interfere with each other.

This animation is embedded in Figure B13 and can be downloaded from (https://github.com/fklich/FSKIDesignAndSupplemental/blob/main/FSKIOrbitAnimationAtL4L5WithTandemPair.mp4) including an audio soundtrack. Click download on right side.

17. SCALABILITY GRID

The grid below in Figure 1 shows how the frequency of the Goldberg polyhedron and n lenses included in the facial plates contributes to FoV coverage.

The n lenses are different for the pentagon faces and the hexagon faces.

The prospective side lengths are shown along with the associated facial plate area to suggest how the lenses could be distributed. Refer to Figure B9.

As shown in Figure 1, the Goldberg polyhedron frequency, the counts of the pentagons and hexagons which will accomodate varieties of n lens layouts suggest the ability for parallel and modular scalabilty.

18. POLYHEDRON DIAMETER, *N* LENSES AND TANDEM MULTIPLES

The polyhedron diameter drives the parallel and modular scalability. The example in this document works with a diameter of 15 feet, akin to the size of a known rocket payload fairing (PLF). This example, factored by the n lenses configured on the polyhedron faces (Figure 1) produces one set of sky coverage factors; this, combined with tandem twin FSKI's produces even higher coverage factors.

Another variant suggests even further factors. If the optical design of the n lenses, their field of view, near infrared (NIR) wavelength capabilities and other parameters could facilitate a lower number of lenses on each of the polyhedron's faces, then the diameter of the sphere might be reduced for efficiency. A smaller sphere could introduce a higher tandem twin, triplet, quadruplet strategy. While this document is for the platform and not optical design, these driving parameters suggest a variety of balancing strategies for higher sky coverage leveraging parallel and modular scalability.

19. HISTORICAL MARKERS FOR *GOING SMALL*

Solutions for improvisation are often successful in history by reductions in size and complexity. Examples are clocks, computers, motors, cameras, batteries, satellites (CubeSats, StarLink). Size reduction improves deployment measures and can also leverage scalability.

20. THE EVOLUTION OF SENSOR DESIGN

This invention is a platform for FSKI, yet its effectiveness will benefit from these improvements in image sensor design.

- Back/Rear illumination
- Stacked CMOS
- Curved focal plane arrays
- Large-Format stitching and tiling

21. MORE THAN ONE POLYHEDRON PAIR

This original design constrained the deployment to one pair of polyhedrons, deployed at L2 and L3, using an intermediate relay satellite. An additional pair of polynedrons can be situated at L4 and L5 locations as suggested in the FSKI progressive series of figures section.

Polyhedron metrics	Goldberg Polyhedron GP{m,n}		
	{2,2}	{3,3}	{4,4}
Total frequency for this Goldberg polyhedron	122	272	482
Full count of pentagons	12	12	12
Full count of hexagons	110	260	470
Truncation at 3/5th (.60)			
net truncated number of pentagons (3/5ths)	7	7	7
net truncated number of hexagons (3/5ths)	80	190	343
Lenses in each pentagon face			
Total lenses using 5 per pentagon	35	35	35
Total lenses using 6 per pentagon	42	42	42
Total lenses using 11 per pentagon	77	77	77
Lenses in each hexagon face			
Total lenses using 7 per hexagon	560	1330	2401
Total lenses using 13 per hexagon	1040	2470	4459
Total lenses using 19 hexagon	1520	3610	6517
Total lenses, combining pentagons + hexagons			
Total lenses (5,7) for entire truncated Polyhedron	595	1365	2436
Total lenses (6,13) for entire truncated Polyhedron	1082	2512	4501
Total lenses (11,19) for entire truncated Polyhedron	1597	3687	6594
Facial areas with hypothetical diameter = 15 ft			
pentagon side length (ft)	2.28	1.52	1.14
pentagon area (sq ft)	8.94	3.96	2.24
hexagon side length (ft)	2.28	1.52	1.14
hexagon area (sq ft)	13.51	5.98	3.38

Figure 1. Goldberg polyhedron metrics of scalability

APPENDIX

A. THE FSKIDESIGNANDSUPPLEMENTAL GITHUB REPOSITORY

Pursue these steps to clone and deploy the constituent files from the GitHub *FSKIDesignAndSupplemental* repository.

- Connect to GitHub repository at https://github.com/fklich/FSKIDesignAndSupplemental
- Clone the complete repository down to a local workstation with unix command *git clone https://github.com/fklich/FSKIDesignAndSupplemental.git* .
- Or, clone the complete *FSKIDesignAndSupplemental* repository by using the green **<>Code .** link and selecting the **Download ZIP** action to receive a zip file on a local workstation. The unzip creates a new local */FSKIDesignAndSupplemental/* directory.
- Now, with the clone complete, the full set of .pdf, .mp4, .jpg files should be present for examination.
- Use **Adobe Acrobat Reader** to open the *FullSky-SphereKinematicsAndImagingV4.pdf* document.
- All the .mp4 files and most of the .jpg files are rendered in the Adobe pdf document. Other PhotoOfxxxxxx.jpg show actual photos of 3D printed prototypes.

B. PROGRESSIVE SERIES OF FIGURES

Figure B1. Full Goldberg polyhedron{3,3} with 12 pentagons and 260 hexagons. This is a design starting point, presenting 272 tapered chambers to house n lenses within each.

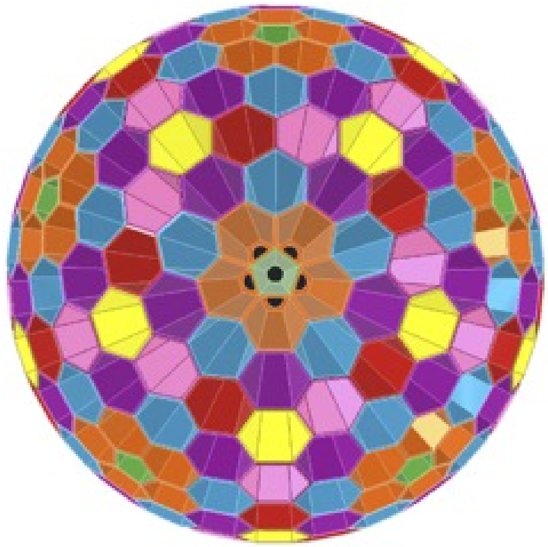

Figure B2. Basic truncated polyhedron. The Full Sky Kinematics-Imaging platform, FSKI, will accommodate 201 lens compartments. Each lens will exist in its individual tapered chamber on a Goldberg polyhedron{3,3}, truncated from a total of 272 faces to 201 faces. As will be illustrated downline in this document, each tapered chamber will accommodate 1 to n lenses. Each lens will aim at a unique point of the celestial sphere. While this platform is intended to operate as an identical coordinated pair, Figure B2 below shows one side profile of the basic polyhedron style of the FSKI.

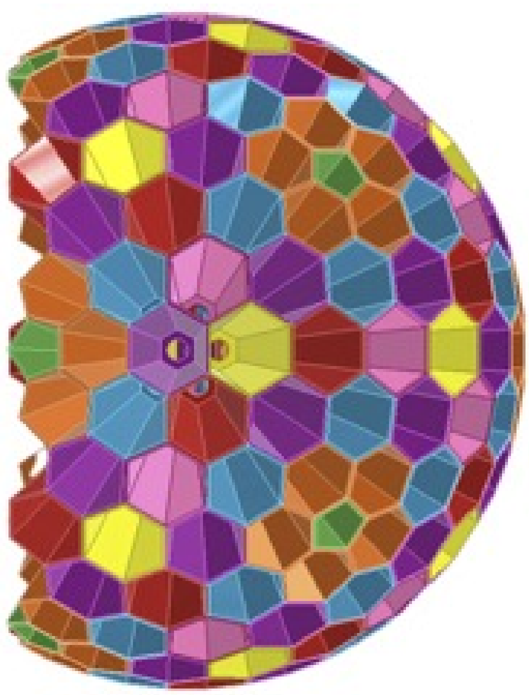

Figure B3. Tandem pair of truncated polyhedrons. These polyhedrons represent the FSKI platform design. They will exist as a complementing pair. FSKI's will support each other by being pointed diametrically opposite in outer space. Figure B3 shows a pair of the FSKI structures as deployed in outer space, each of the pair facing away from the Sun, into $\frac{1}{2}$+ of the full celestial sphere.

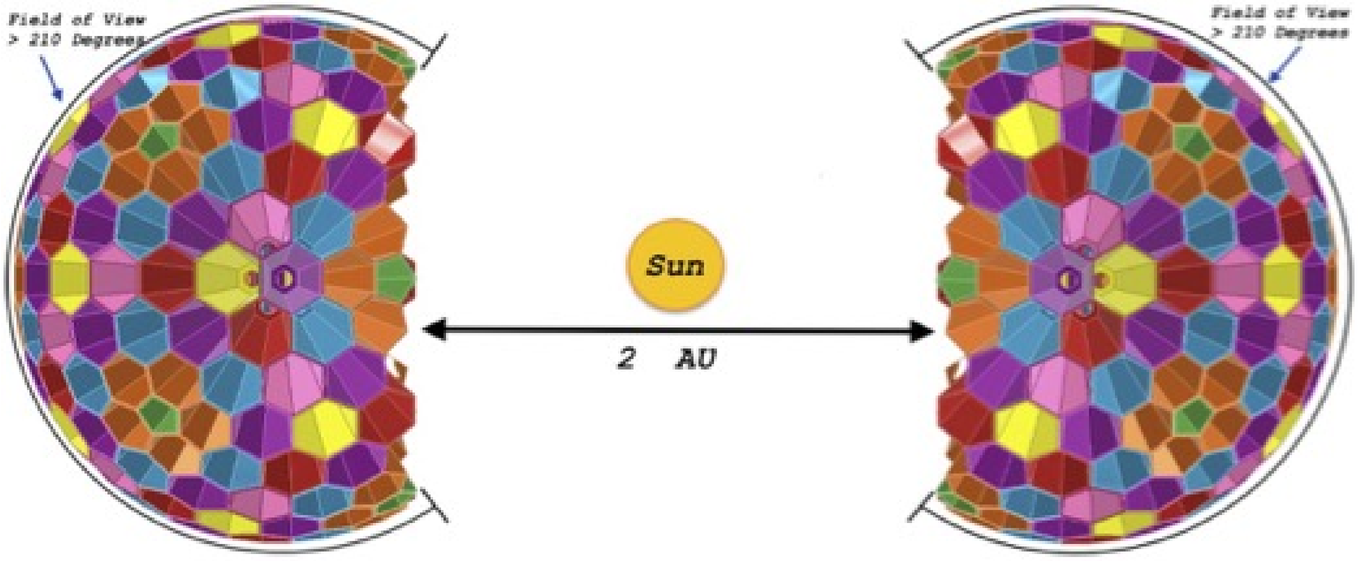

Figure B4. Vacant spaces within the truncated polyhedrons. Each of the two components will include lens foundations that occupy approximately 3/5ths of the full sphere. 2/5ths of the sphere will be open to accommodate other supplementary support components for deployment in outer space. Figure B4 shows the 3/5th spheres rotated at a slight angle revealing the vacated/unoccupied portion of the polyhedron that incorporates a flat base wall.

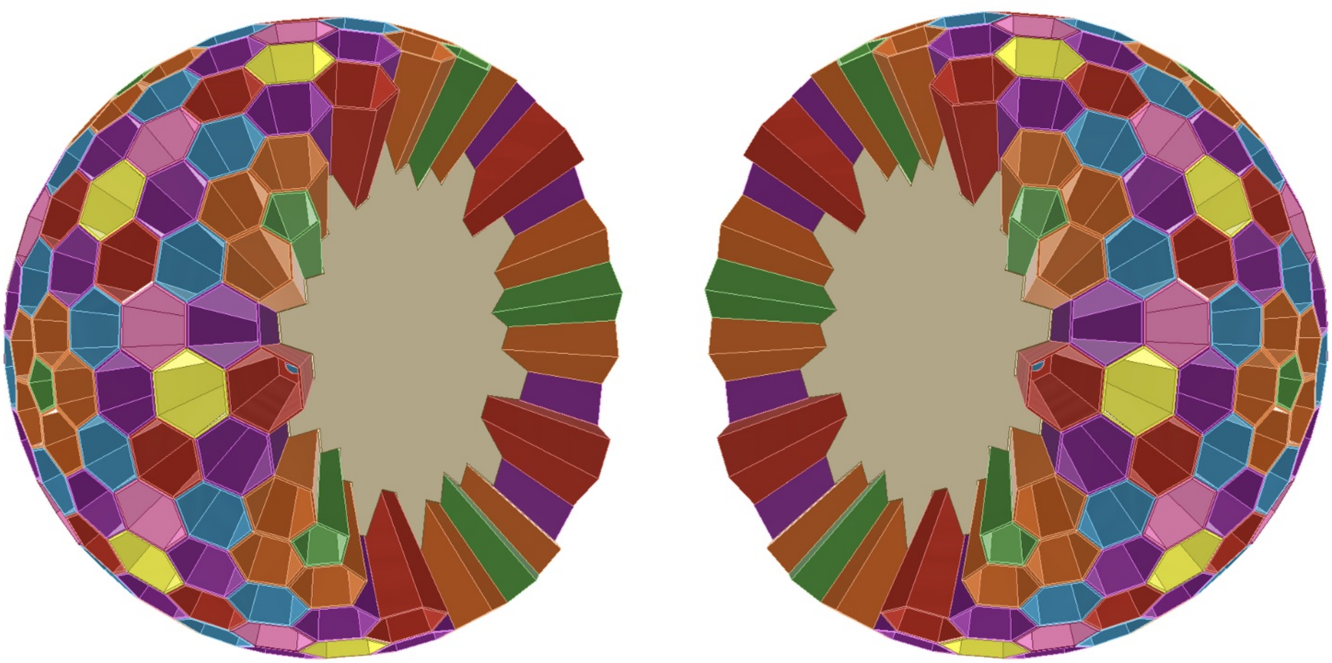

Figure B5. Interior core to serve as equipment bay. Figure B5 shows a common interior core of the polyhedron that will serve as a central equipment bay to enable connectivity from the n lenses in the tapered chambers, the image capture equipment, the packet assembly for outbound transmission and interface to a broadcast radio and a high-gain-antenna (HGA) shown in downline Figure B5. The common core is outlined as a yellow semisphere. The second frame shows a 90 degree rotation and slice with the common core and the holes that lead into the pentagon and hexagon tapered chambers.

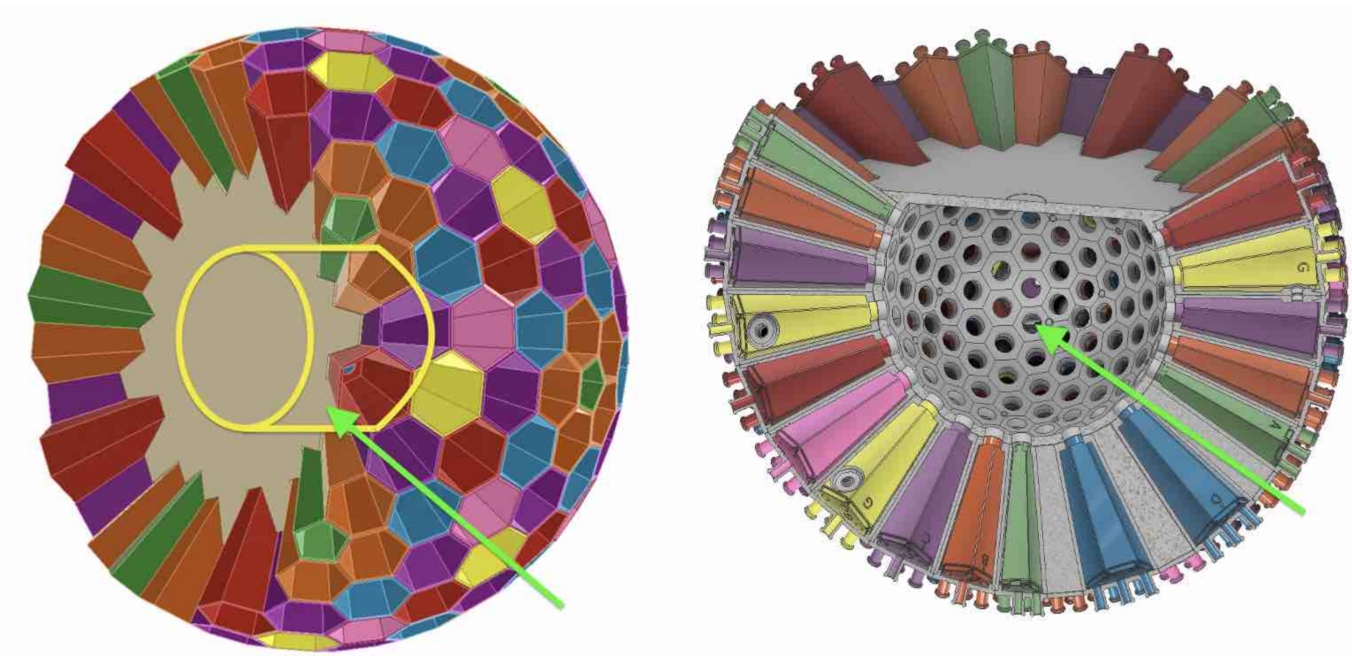

Figure B6. Addition of another pair of basic features to this evolving buildup of diagrams. The architecture includes a central post that will serve to support the umbrella-like sunshield and the high-gain antenna assembly and other equipment.

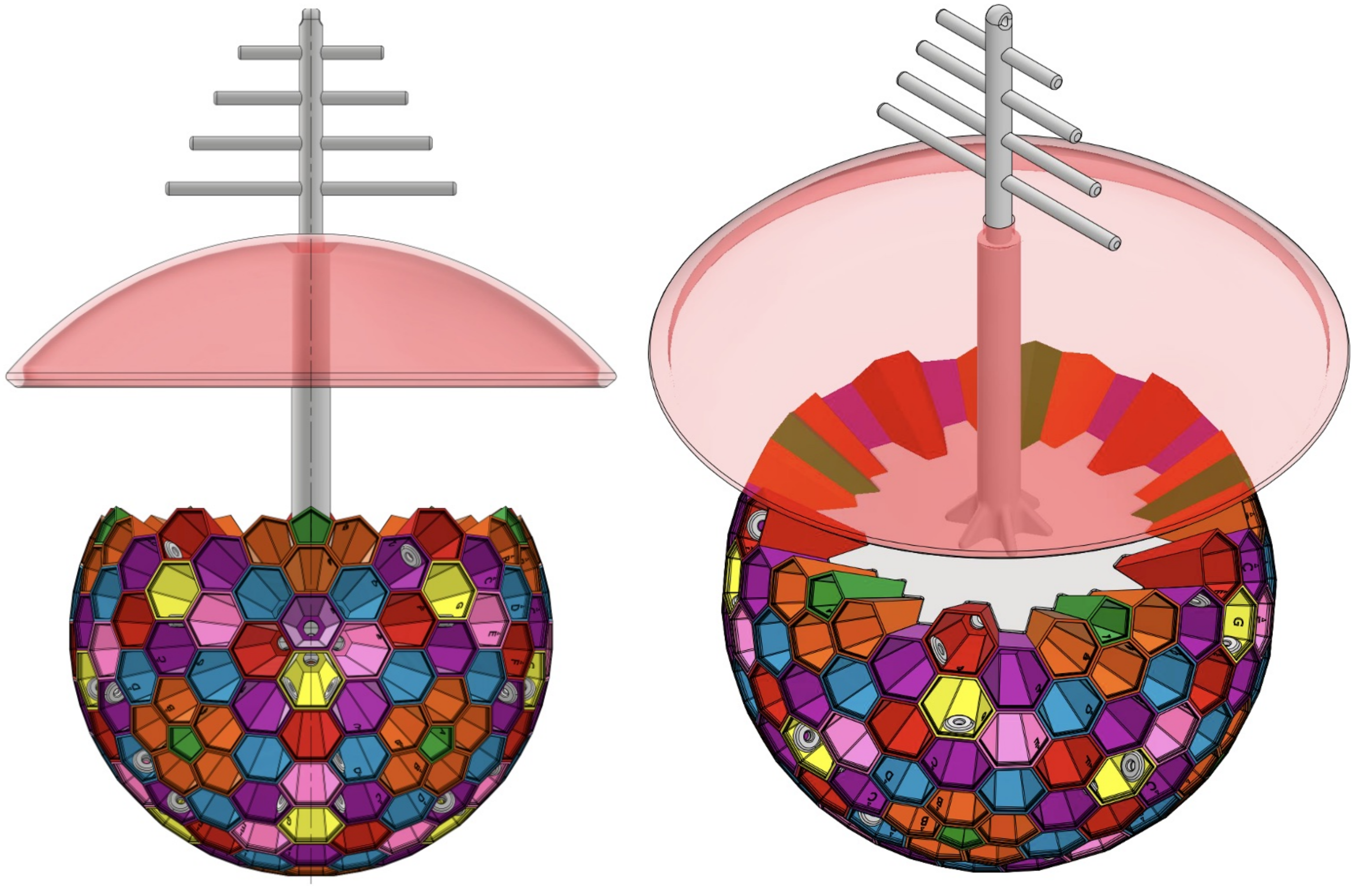

Figure B7. Abbreviated view of tandem pair for deployment in solar orbit. Figure B7 is the next step in the evolving diagrams. Here the full assembly is postured in an abbreviated view of the tandem pair as they will coexist in orbit around the Sun.

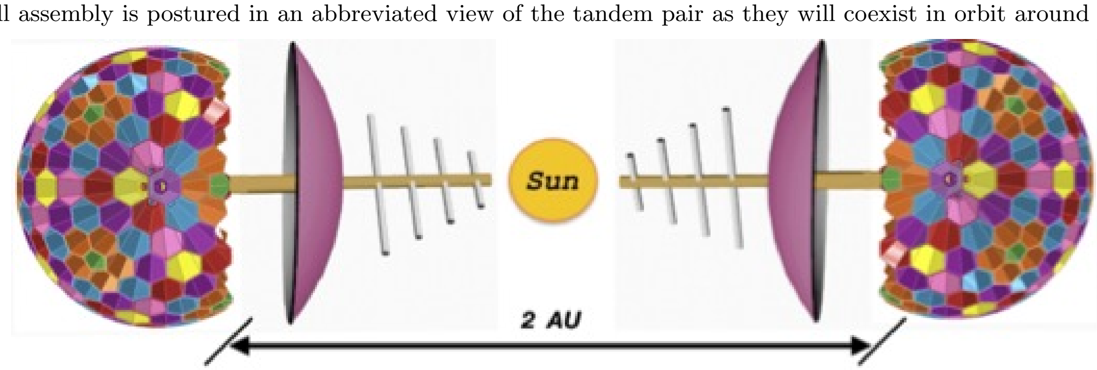

Figure B8. This invention is a platform for FSKI. Figure B8 serves as an example of at least one possible full configuration with n lenses per polyhedron chamber and the total number of lenses for the full truncated polyhedron (N), (i.e. $n = 5$ for pentagons, $n = 7$ for hexagons) in this embodiment.

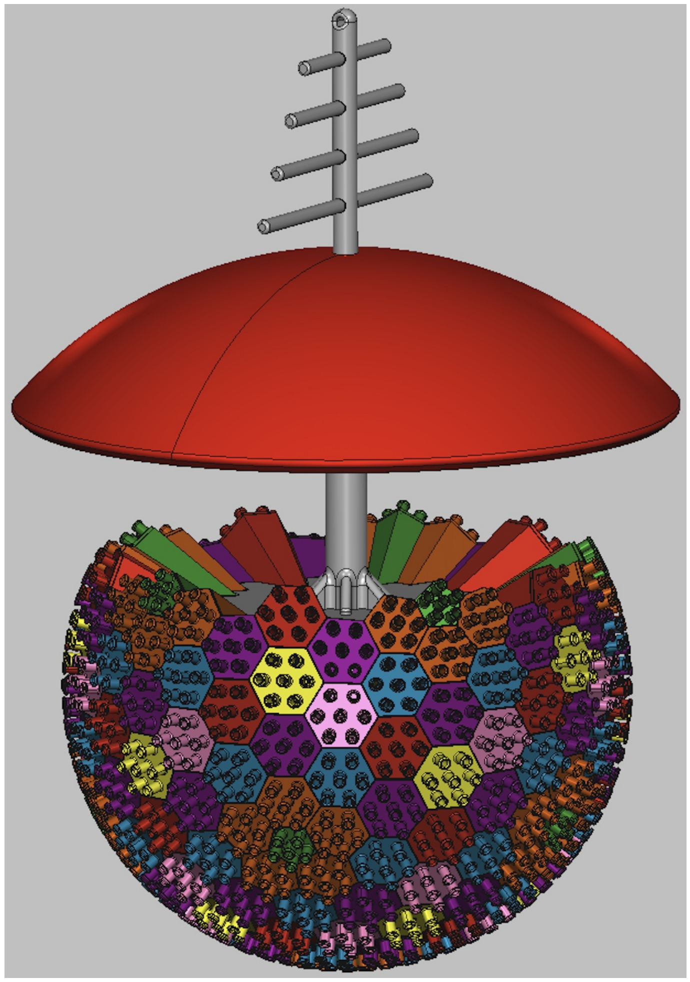

Figure B9. Serves as an embodiment of the lenses configuration on the face of a hexagon. Starting with a single lens within a hexagon with side lengths of 18 inches, frame 2 and 3 show sample arrangement of 7 lenses and 13 lenses.

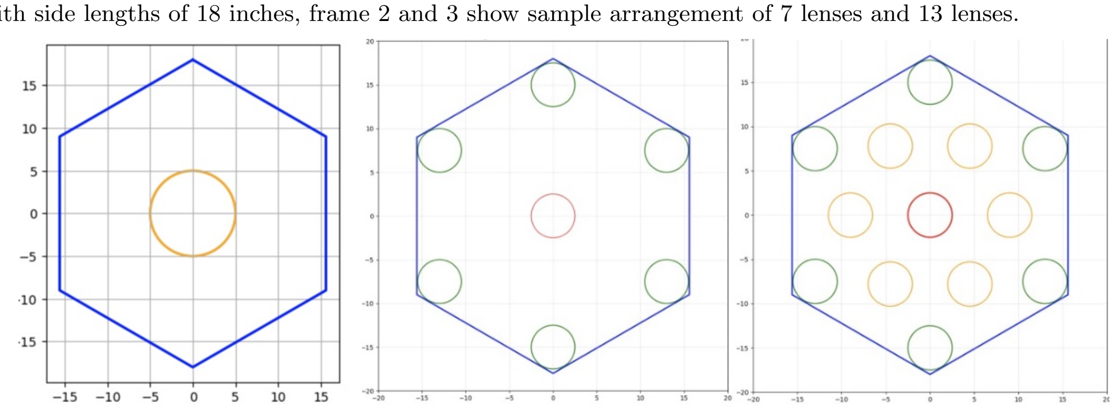

Figure B10. Terrestrial FSKI platform. Figure B10 presents a rather simplistic view of how the FSKI platform could exist in/at a terrestrial locale with all tapered chambers in use.

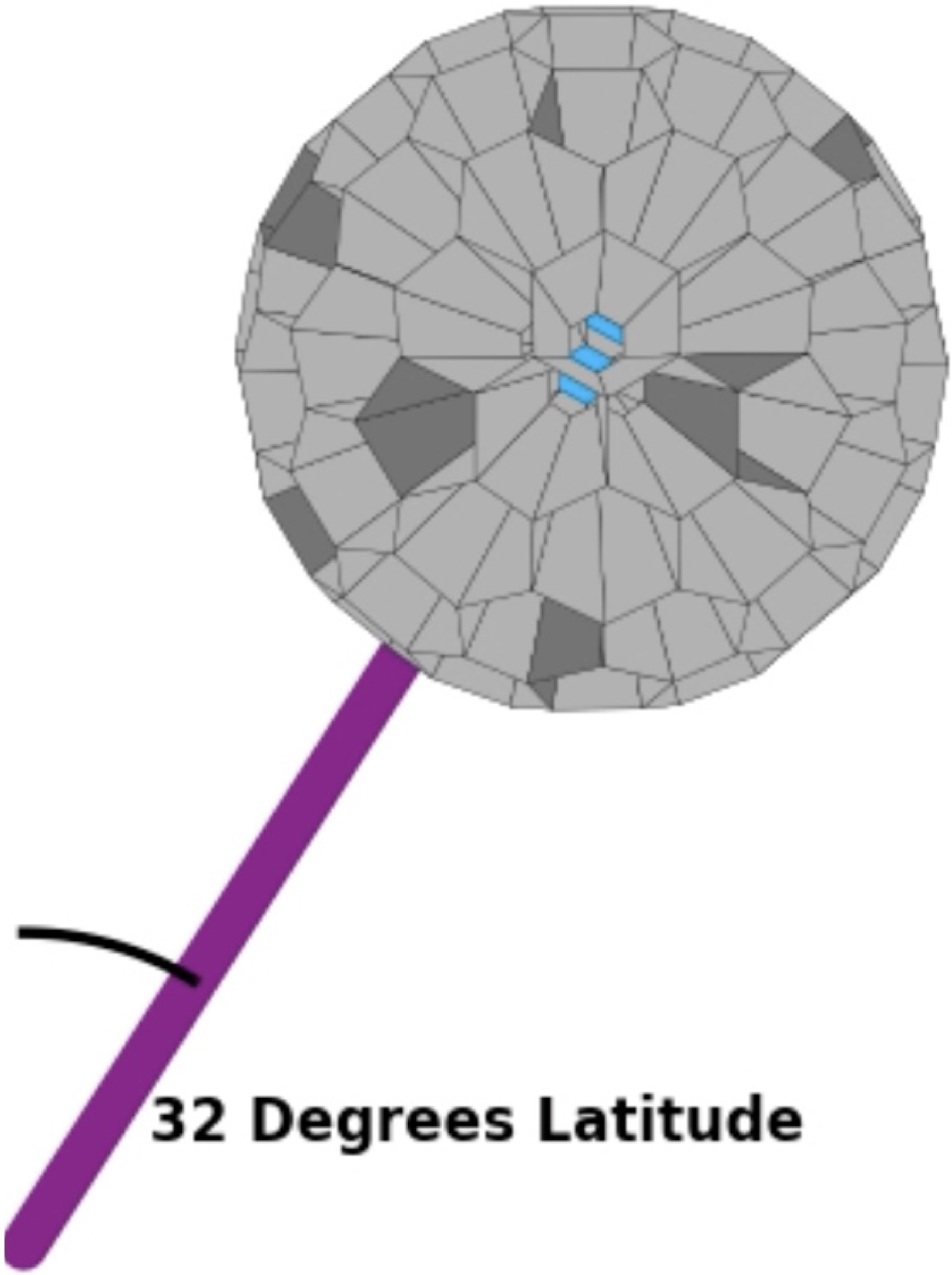

Figure B11. L2, L3, L4 deployment (with animation). The full deployment in outer space of (at least) one pair (2) of these FSKI platforms would leverage the L3, L4 locations in order to sustain (1) opposition and (2) a communications relay to Earth. The top larger frame enhances detail repeated in 6 ensuing frames. The seventh/last miniframe shows overlap in FoV between the two FSKI platforms. Figure B11 reflects a snapshop portrayal of the outer space configuration.

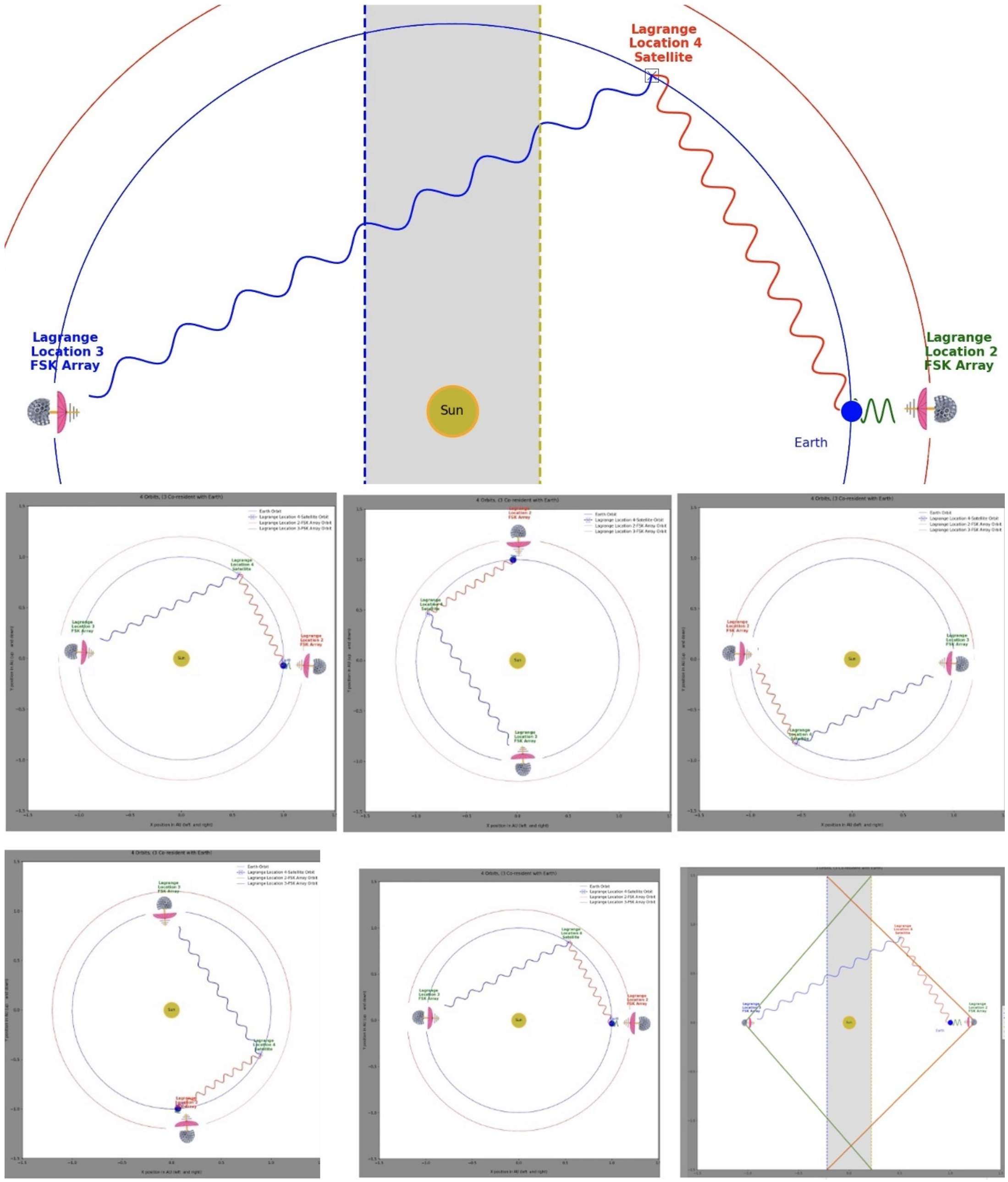

Figure B12. L4, L5 deployment (with animation). Additional L4, L5 deployment with no need for communication relay station to Earth. Figure B12 reflects a static portrayal of the outer space configuration.

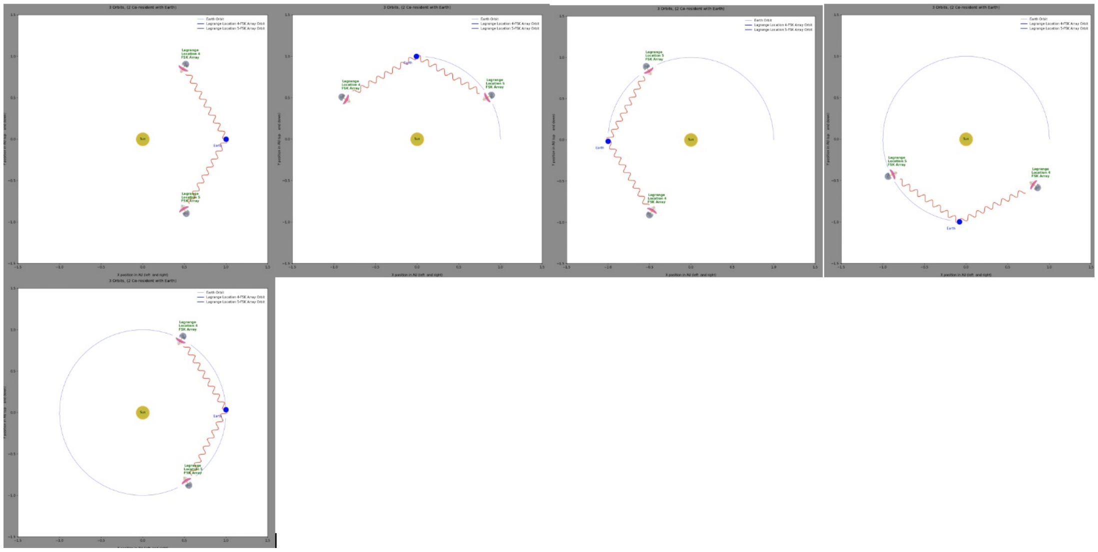

Figure B13. Another embodied L4, L5 deployment (with animation), again, with no need for communication relay station to Earth. Figure B13 reflects a tandem pair at L4 and L5 locations. This configuration would enhance sky coverage by employing a fixed tilt between FSKIa, FSKIb with the same fixed tilt between FSKIc and FSKId.

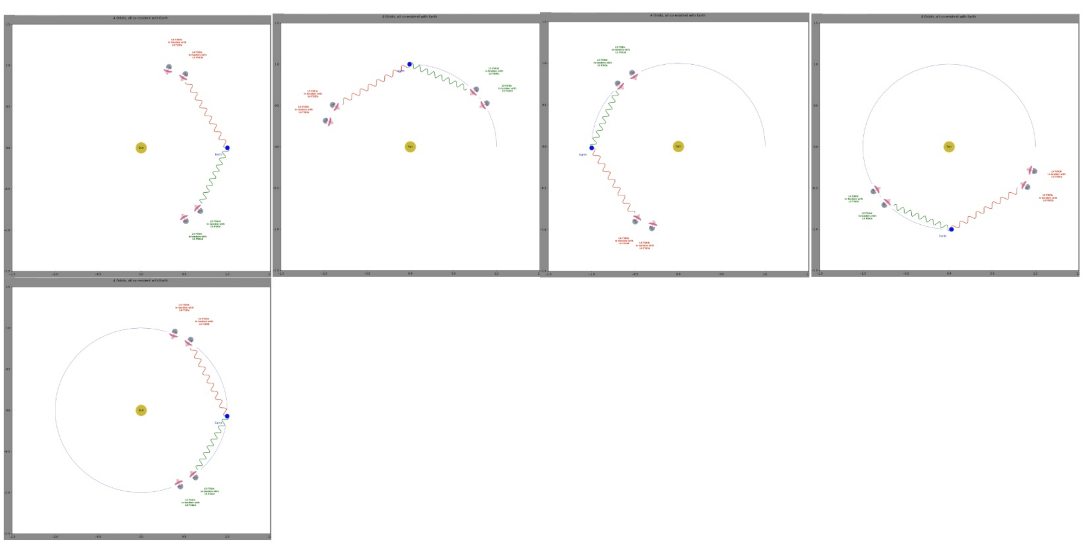

Figure B14. 1/2 sphere sky coverage at 2.50 degrees of target separation. This example represents one FSKI platform FoV with red points indicating FoV overlap with the opposite FSKI unit.

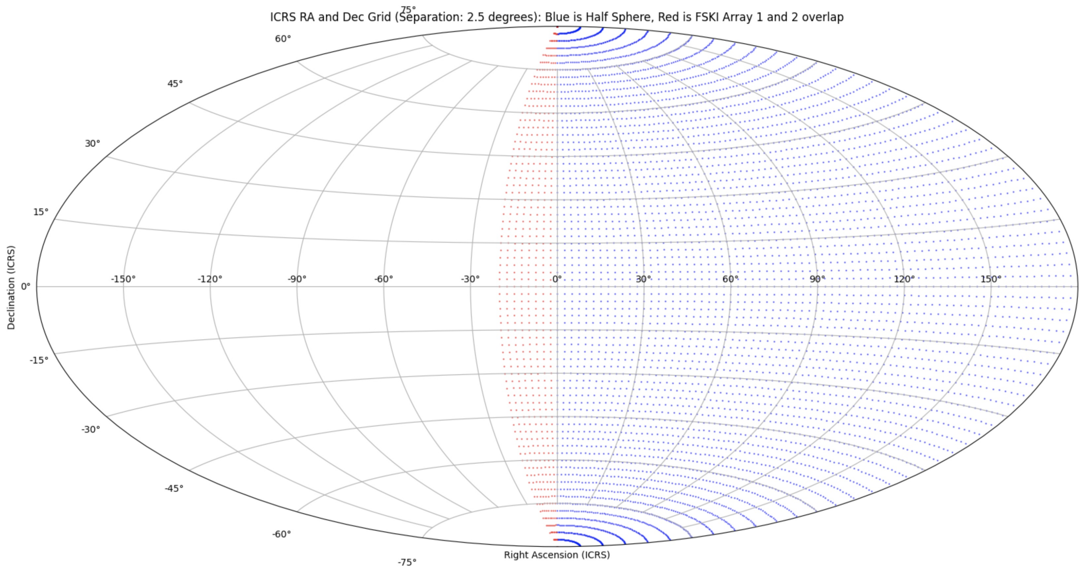

Figure B15. 1/2 sphere sky coverage at 1.25 degrees of target separation, highliting the visual difference in the plotted spacing amongst the points. The same plot can also be made for a target separation of 0.625 with the points becoming far more congested. Again, this example represents one FSKI platform FoV with red points indicating FoV overlap with the opposite FSKI unit.

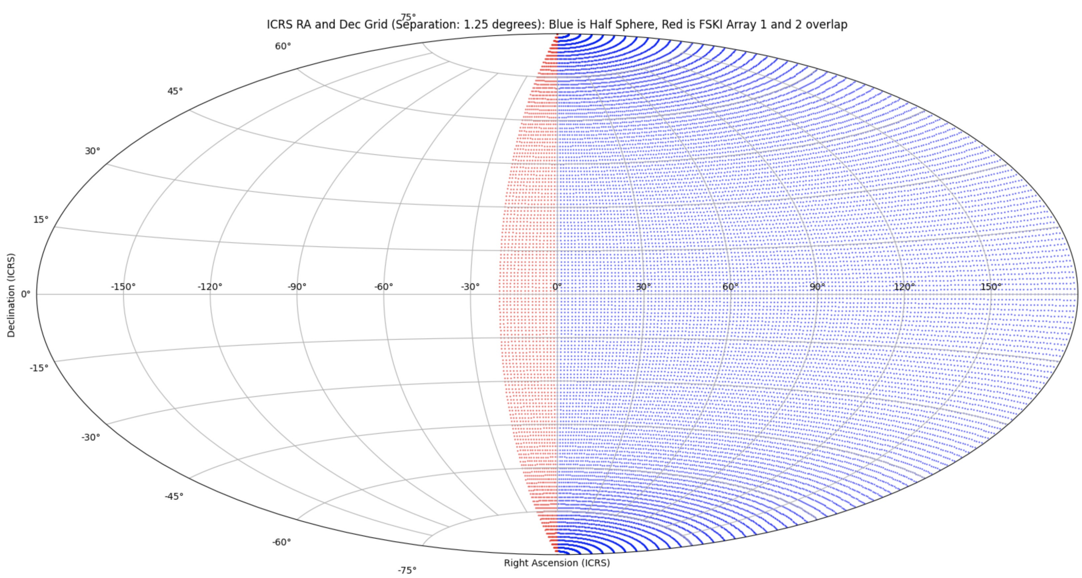

Figure B16. Photo of author, Fred Klich

Figure B17. Photo of 3D print of prototype